CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING (FTE)
DEPARTMENT OF COMPUTER ENGINEERING

MINI-PROJECT ACADEMIC EVALUATION REPORT

MiniBank: Progressive Object-Oriented Banking System & Enterprise Capstone

An Evolutionary Multi-Tier Console Banking Platform Implementing Core and Advanced Java OOP Paradigms (Practicals 01 to 14)

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Certificate of Originality & Submission

This is to certify that the mini-project entitled "MiniBank: Progressive Object-Oriented Banking System & Enterprise Capstone" is a bonafide academic work carried out by Vaidehi Shah (Student ID: D26CE147), student of the 3rd Semester, Department of Computer Engineering, Faculty of Technology and Engineering (FTE), Charotar University of Science and Technology (CHARUSAT), for the course CEUE203: Object-Oriented Programming during the Academic Year 2026-27.

The project has been developed systematically to fulfill all 14 pedagogical milestones of the laboratory curriculum, culminating in a robust, production-grade capstone banking system featuring custom declarative annotations, dynamic reflection, generic repositories, thread-safe concurrent transactions, deadlock avoidance algorithms, binary object serialization, and Java NIO log auditing.

_____	_____
Vaidehi Shah (D26CE147)	Course Faculty / Evaluator
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Acknowledgement

I would like to express my deepest gratitude to our faculty mentors, laboratory instructors, and the Department of Computer Engineering, Faculty of Technology and Engineering (FTE), CHARUSAT, for designing an exceptionally rigorous and rewarding curriculum for CEUE203 (Object-Oriented Programming).

The progressive milestone approach—starting with foundational syntax, records, and encapsulation, moving across inheritance, interface segregation, custom annotations, reflection, multithreading, and culminating in Java NIO audit logging and generic repositories—has provided me with invaluable practical experience in architecting enterprise-grade software in Java (JDK 17+).

Executive Summary

MiniBank is a comprehensive, semester-long software engineering endeavor conceived to demonstrate the practical application of Object-Oriented Programming (OOP) principles and modern Java language paradigms. Rather than developing an isolated prototype, the project is structured along two synergistic dimensions:

1. Progressive Milestone Evolution (Practicals 01 through 14): Fourteen distinct, standalone milestone stages that track the student's laboratory journey, demonstrating how a console banking shell incrementally incorporates encapsulation, immutability, pattern matching, polymorphism, interface segregation, custom checked exceptions, multithreaded synchronization, deadlock prevention, binary

serialization, collections, and generic persistence.

2. The Enterprise Capstone Application (minibank package): An integrated, layered architecture that unifies all fourteen practical milestones into a clean, production-grade banking engine. The capstone features a decoupled design with a Model layer, Annotation validation layer, Custom Exception hierarchy, Generic In-Memory Repository with reflection-driven primary key detection, Thread-safe Transaction Processor with Dijkstra-style resource ordering to prevent deadlocks, Binary State Persistence (accounts.dat), Java NIO Files DirectoryStream Audit Report Generator, and an automated integration test suite verifying 100% test pass rate.

> 📌 **SUBMISSION & VERIFICATION DETAILS:** Complete source code repository: https://github.com/d26ce147-prog/D26CE147_OOP_MiniProject.git
> Assignment Submission Form: https://forms.gle/qh3yMuouwKyGLdwL6
> Target Execution Platform: Java SE 17+ (LTS) / 21+ on Windows / Linux / macOS

Chapter 1: Introduction & Project Scope

1.1 Course Context & Pedagogical Scope

The course CEUE203 (Object-Oriented Programming) at FTE, CHARUSAT, aims to transition computer engineering students from basic procedural coding to industrial-grade software engineering using Java. Modern banking systems provide an ideal pedagogical domain because they mandate absolute state consistency, strict data encapsulation, rigorous input validation, granular exception handling, deterministic concurrency without deadlocks, and verifiable audit trails.

1.2 Problem Statement & Domain Requirements

Traditional classroom exercises often isolate OOP concepts into disconnected toy examples (e.g., shapes, employee payrolls). In contrast, modern banking demands that multiple complex concepts work synchronously in a high-concurrency, failure-sensitive environment:

• State Invariants: Account balances must never fall below minimum required thresholds or overdraft limits.

• Type Polymorphism: Savings, Current, and Fixed Deposit accounts exhibit vastly different rules for interest accrual and withdrawal allowances.

• Concurrency & Deadlocks: Simultaneous fund transfers between cross-locking accounts (Account A transferring to B while B transfers to A) can easily lead to deadlock conditions if locks are acquired arbitrarily.

• Resilience & Auditing: Transactions must be logged atomically to disk, and historical operations must be audited via fast, modern stream I/O.

1.3 Project Architecture Overview

MiniBank addresses these challenges by dividing the project into a dual-structure repository:

1. Milestone Directory (Practical_01 through Practical_14): Enables academic reviewers to verify the step-by-step curriculum progression.

2. Enterprise Capstone Architecture (Full_MiniBank_Application): A production-grade multi-package Java system structured into clean architectural boundaries:

- minibank.model: Domain entities and record structures.
- minibank.model.annotation: Custom runtime metadata annotations (@Id, @Positive, @MaxLength).
- minibank.interfaces: Behavioral contracts (Transactable, InterestBearing, Premium, WithdrawRule).
- minibank.exception: Checked exception hierarchy with shortfall computation.
- minibank.util: Regex validator, annotation reflection validator, statement formatter, command parser.
- minibank.repository: Generic Repository<T> utilizing Java Generics and reflection.
- minibank.service: BankService orchestrator, TransactionProcessor, StatePersister, ReportGenerator, and TransactionLog.
- minibank.app: Interactive console user interface (MiniBank) and automated unit test suite (MiniBankTests).

Directory / Module	Core Focus	Architectural Contribution
Practical_01	Class Fundamentals & Shell	BankInfo record, MenuOption enum, Switch expressions, interactive loop.
Practical_02	Encapsulation & State Invariants	Customer model, private state, invariant checks, deposit/withdraw guards.
Practical_03	Object Methods & Static Nested	toString, equals, hashCode on account IDs; static nested Address class.
Practical_04	Regex Validation & Command Parser	Validator regex for mobile, email, PAN, IFSC; Command record parser.
Practical_05	Inheritance & Polymorphism	Abstract Account; SavingsAccount, CurrentAccount, FixedDepositAccount.
Practical_06	Interfaces & Lambda Rules	Transactable, InterestBearing, Premium marker; functional WithdrawRule.
Practical_07	Annotations & Reflection	Custom @Id, @Positive, @MaxLength; runtime reflection validation engine.
Practical_08	Custom Checked Exceptions	BankException hierarchy; InsufficientFundsException with shortfall.
Practical_09	Thread Synchronization	Synchronized balance operations; concurrent AccountWorker threads.
Practical_10	Executor Pools & Deadlock Guard	Deterministic resource locking algorithm to prevent circular waits.
Practical_11	Persistence & Java NIO Audit	Binary ObjectOutputStream serialization; Java NIO log stream analyzer.
Practical_12	Collections & Sorting	Map-backed account index; Comparable and Comparator balance sorting.
Practical_13	Generic Repository Pattern	Repository<T> storing any entity with reflection-based @Id extraction.
Practical_14	Capstone Assembly & Suite	Full BankService coordinator, automated test runner with 100% pass rate.
Full_MiniBank_Application	Modular Enterprise Capstone	Complete 8-package production codebase with tests, logs, and data.

Chapter 2: System Architecture & Design

2.1 Layered Architectural Diagram

MiniBank adopts a strict, decoupled Layered Architecture following industry-standard separation of concerns. Dependencies flow strictly downward: UI / Tests -> Service Layer -> Repository Layer -> Domain Model & Utilities.

Listing: Layered System Architecture of MiniBank Capstone

```
+-------------------------------------------------------------------------------+
|                            PRESENTATION LAYER (CLI)                           |
|      MiniBank (Interactive CLI Menu)      |      MiniBankTests (Automated Suite)  |
+-----------------------------------------+-----------------------------------------+
                                          | invokes
                                          v
+-------------------------------------------------------------------------------+
|                                 SERVICE LAYER                                 |
|  BankService (Coordinator)               |  TransactionProcessor (ExecutorService)|
|  StatePersister (Object Serialization)   |  ReportGenerator (Java NIO Directory)  |
+-----------------------------------------+-----------------------------------------+
                                          | delegates
                                          v
+-------------------------------------------------------------------------------+
|                            DATA ACCESS / REPOSITORY                           |
|  Repository<T> (Generic CRUD Engine)     |  AccountStore (Concurrent Map Storage) |
+-----------------------------------------+-----------------------------------------+
                                          | manages
                                          v
+-------------------------------------------------------------------------------+
|                               DOMAIN MODEL LAYER                              |
|  Account (Abstract Base)  <|--- SavingsAccount (Min Balance, 4% Interest)         |
|                           <|--- CurrentAccount (Overdraft Limit, 0% Interest)     |
|                           <|--- FixedDepositAccount (Lock Policy, 7% Interest)    |
|  Customer (Nested Address)|  BankInfo (Record)   |  Command (Parser Record)       |
+-------------------------------------------------------------------------------+
|                    CROSS-CUTTING UTILITIES & METADATA INFRASTRUCTURE             |
|  @Id, @Positive, @MaxLength              |  AnnotationValidator (Reflection)       |
|  Validator (Regex Mobile/Email/PAN/IFSC)|  BankException Hierarchy (Shortfall)    |
+-------------------------------------------------------------------------------+
```

2.2 Domain Model & Polymorphic Hierarchy

At the foundation of MiniBank is the abstract class Account, which defines the state and invariants common to all bank accounts: accountNumber, ownerName, and balance. It implements three core interfaces: Transactable, InterestBearing, and Comparable<Account>, as well as the Serializable marker interface for persistent disk storage.

Polymorphism is cleanly leveraged across three specialized account variants:
• SavingsAccount: Enforces a minimum balance rule (minBalance). Any withdrawal that would reduce the balance below this threshold is rejected. Accrues 4.0% annual interest.
• CurrentAccount: Permits negative balances up to an agreed overdraftLimit (e.g., Rs. 5000). Commercial accounts accrue 0.0% interest.

• FixedDepositAccount: Locked term deposit offering 7.0% annual interest. The canWithdraw() guard strictly returns false to protect principal.

2.3 Interface Segregation & Behavioral Contracts

In conformance with the Interface Segregation Principle (ISP), MiniBank defines concise, focused interfaces:
• Transactable: Defines core transactional contracts (deposit, withdraw, getBalance).
• InterestBearing: Specifies double interestRate() calculation.
• Premium: A marker interface used to tag high-value accounts eligible for priority processing.
• WithdrawRule: A functional interface (@FunctionalInterface) defining boolean allow(Account acc, long amount), enabling lambda-based custom withdrawal policies without altering the account class.

2.4 Generic In-Memory Repository with Reflection

Instead of hardcoding storage for accounts, MiniBank implements a completely reusable, type-safe generic repository class: Repository<T>. The repository maintains an internal ConcurrentHashMap<String, T>. When saving an entity via save(T entity), the repository traverses the entity's class hierarchy via Java Reflection, locates the field annotated with @Id, sets accessible permissions, extracts the string identifier, and uses it as the map key. This decouples storage logic from entity schema.

2.5 Deadlock Prevention in Concurrent Transfers

A notorious pitfall in multithreaded banking is the dining philosophers / deadlock problem during two-party fund transfers. If Thread 1 executes transfer(A, B) and Thread 2 executes transfer(B, A), Thread 1 acquires Lock(A) and waits for Lock(B), while Thread 2 acquires Lock(B) and waits for Lock(A)—producing an unrecoverable circular wait deadlock.

MiniBank solves this elegantly in TransactionProcessor.transferDeadlockFree(Account a, Account b, long amount) by establishing a deterministic global lock order using lexicographical account number comparison (a.getAccountNumber().compareTo(b.getAccountNumber()) < 0). Both threads are guaranteed to acquire the locks in identical sequence (First lock Account A, then Account B), breaking Dijkstra's circular wait condition.

2.6 Persistence & Modern Java NIO Audit Logging

System reliability requires durable state persistence and compliance auditing:
• Binary Object Serialization (StatePersister): On application shutdown (Option 8), all account entities are serialized to data/accounts.dat using ObjectOutputStream. On restart, ObjectInputStream deserializes the exact account states without data loss.
• Java NIO File Audit Stream (ReportGenerator): Financial operations are logged to logs/transactions.log. The ReportGenerator uses Java NIO (Files.newDirectoryStream, Files.readAttributes, Files.newBufferedReader) to scan the log directory, inspect timestamps and file sizes, stream transaction records, compute total deposits, total withdrawals, and net cash movement.

Chapter 3: Curricular Milestone Breakdown (Practicals 01 to 14)

This chapter documents how MiniBank evolved across all 14 curricular milestones defined in the CEUE203 syllabus. Each practical addresses a distinct, foundational concept of Object-Oriented Programming and modern Java architecture.

Practical 01: Class Fundamentals, Console Menu & Records

Core OOP Concept: Java 17+ Records, Enums, Scanner Input, Looping Switch Expressions

Practical 01 lays the foundation for the banking shell. It defines BankInfo as an immutable record to hold read-only metadata (bank name and campus branch), an enum MenuOption with numeric codes, and an interactive CLI loop powered by modern Java switch expressions.

Listing: Practical 01 Representative Implementation

```
record BankInfo(String name, String branch) {}
enum MenuOption {
    OPEN_ACCOUNT(1, "Open Account"), DEPOSIT(2, "Deposit Funds"), ...;
    public static MenuOption fromChoice(int ch) { ... }
}
```

Practical 02: Encapsulation, State Invariants & Customer Entity

Core OOP Concept: Data Hiding, Private Fields, Mutator Guards, Balance Invariants

Practical 02 models the Customer and Account domain entities. All member variables (balance, ownerName, accountNumber) are marked private. Mutator methods enforce state integrity: deposits must be strictly positive, and withdrawals cannot exceed the available balance.

Listing: Practical 02 Representative Implementation

```
public void deposit(long amt) {
    if (amt <= 0) throw new IllegalArgumentException("Amount must be > 0");
    this.balance += amt;
}
public boolean withdraw(long amt) {
    if (amt > 0 && this.balance >= amt) { this.balance -= amt; return true; }
    return false;
}
```

Practical 03: Object Methods & Static Nested Classes

Core OOP Concept: Overriding toString, equals, hashCode; Static Nested Address Class; Cloneable

Practical 03 overrides java.lang.Object methods. Two accounts are defined as equal if and only if their unique account numbers match, maintaining consistent hashCode contracts. Customer embeds a static nested Address class with street, city, and zip code.

Listing: Practical 03 Representative Implementation

```
@Override
public boolean equals(Object o) {
    if (this == o) return true;
    if (!(o instanceof Account a)) return false;
    return Objects.equals(this.accountNumber, a.accountNumber);
}
@Override
public int hashCode() { return Objects.hash(accountNumber); }
```

Practical 04: String Processing, Regex Validation & Command Records

Core OOP Concept: Regex Pattern Matching, Tokenization, Command Parsing Record

Practical 04 implements a robust validation subsystem (Validator) utilizing compiled java.util.regex.Pattern objects for Indian mobile numbers (^[6-9]\d{9}$), email addresses, PAN cards, and IFSC codes. A Command record parses console input into structured action tokens.

Listing: Practical 04 Representative Implementation

```
public class Validator {
    private static final Pattern MOBILE_PATTERN = Pattern.compile("^[6-9]\\d{9}$");
    public static boolean isValidMobile(String m) {
        return m != null && MOBILE_PATTERN.matcher(m).matches();
    }
}
```

Practical 05: Class Inheritance & Polymorphism

Core OOP Concept: Abstract Base Classes, Method Overriding, Polymorphic Dispatch

Practical 05 introduces an abstract Account base class with specialized subclasses: SavingsAccount (enforcing minimum balance), CurrentAccount (supporting overdraft credit), and FixedDepositAccount (locked funds). Polymorphic dispatch allows the banking service to process accounts uniformly.

Listing: Practical 05 Representative Implementation

```
public abstract class Account {
    public abstract double interestRate();
    public abstract boolean canWithdraw(long amount);
}
public class SavingsAccount extends Account {
    @Override public double interestRate() { return 4.0; }
    @Override public boolean canWithdraw(long amt) { return (balance - amt) >= minBalance; }
}
```

Practical 06: Interfaces, Markers & Lambda Rules

Core OOP Concept: Interface Segregation, Marker Interfaces, Functional Interfaces & Lambdas

Practical 06 implements Transactable, InterestBearing, and Premium marker interfaces. A functional interface WithdrawRule (@FunctionalInterface) allows callers to inject dynamic withdrawal rules using concise Java lambda expressions.

Listing: Practical 06 Representative Implementation

```
@FunctionalInterface
public interface WithdrawRule {
    boolean allow(Account account, long amount);
}
// In caller:
WithdrawRule vipRule = (acc, amt) -> acc.getBalance() >= amt;
if (vipRule.allow(acc, amount)) acc.withdraw(amount);
```

Practical 07: Custom Metadata Annotations & Reflection

Core OOP Concept: Declarative Programming, Custom Annotations, Java Reflection API

Practical 07 introduces custom runtime annotations (@Id, @Positive, @MaxLength) applied to entity fields. A reflection-based AnnotationValidator inspects declared fields at runtime, verifies constraints, and returns human-readable validation error messages.

Listing: Practical 07 Representative Implementation

```
@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.FIELD)
public @interface Positive { String message() default "Must be positive"; }
// Runtime inspection via Class.getDeclaredFields() and Field.getAnnotation()
```

Practical 08: Custom Checked Exception Hierarchy

Core OOP Concept: Checked Exceptions, Shortfall Tracking, Fault-Tolerant Error Handling

Practical 08 replaces silent boolean failure returns with an expressive checked exception hierarchy: BankException, InsufficientFundsException (which records the exact calculated monetary shortfall), InvalidAmountException, and AccountNotFoundException.

Listing: Practical 08 Representative Implementation

```
public class InsufficientFundsException extends BankException {
    private final long shortfall;
    public InsufficientFundsException(String msg, long shortfall) {
        super(msg); this.shortfall = shortfall;
    }
    public long getShortfall() { return shortfall; }
}
```

Practical 09: Concurrency & Thread Synchronization

Core OOP Concept: Multithreading, Thread Safety, Synchronized Critical Sections

Practical 09 demonstrates thread safety in high-concurrency environments. Mutating operations (deposit, withdraw) are marked synchronized to prevent race conditions when multiple AccountWorker threads concurrently manipulate shared account balances.

Listing: Practical 09 Representative Implementation

```
public synchronized void deposit(long amount) throws InvalidAmountException {
    if (amount <= 0) throw new InvalidAmountException("Amount must be > 0");
    this.balance += amount;
}
```

Practical 10: Managed Thread Pools & Deadlock Avoidance

Core OOP Concept: ExecutorService, Deterministic Resource Ordering, Deadlock Prevention

Practical 10 introduces a managed thread pool (Executors.newFixedThreadPool) and implements Dijkstra's resource ordering in transferDeadlockFree(). By ordering account locks lexicographically based on account number, cross-transfer circular wait deadlocks are mathematically eliminated.

Listing: Practical 10 Representative Implementation

```
public static void transferDeadlockFree(Account a, Account b, long amount) throws BankException
{
    Account first = a.getAccountNumber().compareTo(b.getAccountNumber()) < 0 ? a : b;
    Account second = (first == a) ? b : a;
    synchronized (first) { synchronized (second) {
        a.withdraw(amount); b.deposit(amount);
    }}
}
```

Practical 11: Object Serialization & Java NIO Log Auditing

Core OOP Concept: Binary ObjectOutputStream, Java NIO Path, DirectoryStream, File Attributes

Practical 11 delivers full persistence and audit capability. Accounts are saved to data/accounts.dat via ObjectOutputStream. A ReportGenerator class walks log directories via Files.newDirectoryStream("*.log"), reading file attributes and aggregating deposit/withdrawal totals.

Listing: Practical 11 Representative Implementation

```
try (DirectoryStream<Path> stream = Files.newDirectoryStream(logDir, "*.log")) {
    for (Path p : stream) {
        BasicFileAttributes attrs = Files.readAttributes(p, BasicFileAttributes.class);
        // parse deposit and withdrawal lines...
    }
}
```

Practical 12: Java Collections Framework & Account Sorting

Core OOP Concept: Map Lookups, Comparable Natural Ordering, Comparator Strategy

Practical 12 transitions internal storage from fixed arrays to dynamic Java Collections. Accounts are indexed in a ConcurrentHashMap for O(1) lookups. Sorting by account number uses Comparable<Account>, while sorting by balance utilizes a dedicated Comparator<Account>.

Listing: Practical 12 Representative Implementation

```
public class AccountStore {
    private final Map<String, Account> accounts = new ConcurrentHashMap<>();
    public List<Account> getSortedByBalance() {
        return accounts.values().stream()
            .sorted(Comparator.comparingLong(Account::getBalance).reversed())
            .toList();
    }
}
```

Practical 13: Generic Repository Pattern

Core OOP Concept: Java Generics (T), Bounded Type Parameters, Dynamic Entity Persistence

Practical 13 designs a fully generic Repository<T> capable of managing any domain entity. It incorporates generic CRUD operations (save, findById, findAll, delete, count) and automatically identifies primary key fields annotated with @Id using reflection.

Listing: Practical 13 Representative Implementation

```
public class Repository<T> {
    private final Map<String, T> storage = new ConcurrentHashMap<>();
    public T save(T entity) {
        String key = extractId(entity);
        storage.put(key, entity);
        return entity;
    }
}
```

Practical 14: Capstone Assembly & Automated Test Verification

Core OOP Concept: System Integration, Layered Architecture, Automated Testing Harness

Practical 14 unifies all previous 13 milestones into a complete, end-to-end banking application. The BankService coordinates transactions, persistence, and auditing. An automated test harness (MiniBankTests) verifies business logic, exceptions, shortfall calculations, and concurrent batch execution.

Listing: Practical 14 Representative Implementation

```
public static void main(String[] args) {
    BankService service = new BankService();
    // Automated verification of deposit, overdraft shortfall, invalid input, and concurrency
    System.out.printf("MINIPROJECT TEST SUMMARY: %d / %d TESTS PASSED\n", passed, total);
}
```

Chapter 4: Implementation Details & Key Innovations

4.1 Custom Declarative Annotations & Reflection Engine

To decouple business validation from core model logic, MiniBank creates a declarative validation framework. Domain models declare constraints via custom annotations:

- @Id: Marks the unique identity property of an entity.
- @Positive: Ensures numeric balances or amounts are strictly greater than zero.
- @MaxLength: Constrains string lengths (e.g., account holder names or branch descriptions).

The AnnotationValidator inspects declared fields at runtime using Reflection, even on private fields via setAccessible(true).

Listing: AnnotationValidator: Runtime Reflection Validation Engine

```
package minibank.util;
import minibank.model.annotation.Positive;
import minibank.model.annotation.MaxLength;
import java.lang.reflect.Field;
import java.util.ArrayList;
import java.util.List;

public class AnnotationValidator {
    public static String[] validate(Object obj) {
        List<String> errors = new ArrayList<>();
        Class<?> clazz = obj.getClass();
        for (Field f : clazz.getDeclaredFields()) {
            f.setAccessible(true);
            try {
                if (f.isAnnotationPresent(Positive.class)) {
                    Object val = f.get(obj);
                    if (val instanceof Number n && n.doubleValue() <= 0) {
                        Positive p = f.getAnnotation(Positive.class);
                        errors.add(f.getName() + ": " + p.message());
                    }
                }
                if (f.isAnnotationPresent(MaxLength.class)) {
                    Object val = f.get(obj);
                    if (val != null && val.toString().length() >
f.getAnnotation(MaxLength.class).value()) {
                        errors.add(f.getName() + " length exceeds limit");
                    }
                }
            } catch (IllegalAccessException ignored) {}
        }
        return errors.toArray(new String[0]);
    }
}
```

4.2 Generic In-Memory Repository with Reflection-Driven ID Resolution

Repository<T> provides high-level CRUD capabilities for any entity type. The method extractId(T entity) dynamically walks the inheritance tree to locate any field tagged with @Id, eliminating repetitive key-extraction boilerplate.

Listing: Generic Repository<T> with Dynamic Primary Key Extraction

```
package minibank.repository;
import minibank.model.annotation.Id;
import java.lang.reflect.Field;
import java.util.*;
import java.util.concurrent.ConcurrentHashMap;

public class Repository<T> {
    private final Map<String, T> storage = new ConcurrentHashMap<>();

    public T save(T entity) {
        String key = extractId(entity);
        storage.put(key, entity);
        return entity;
    }
    public T findById(String id) { return storage.get(id); }
    public List<T> findAll() { return new ArrayList<>(storage.values()); }
    public boolean delete(String id) { return storage.remove(id) != null; }
    public int count() { return storage.size(); }

    private String extractId(T entity) {
        Class<?> curr = entity.getClass();
        while (curr != null && curr != Object.class) {
            for (Field f : curr.getDeclaredFields()) {
                if (f.isAnnotationPresent(Id.class)) {
                    f.setAccessible(true);
                    try { return String.valueOf(f.get(entity)); }
                    catch (IllegalAccessException e) { throw new RuntimeException(e); }
                }
            }
            curr = curr.getSuperclass();
        }
        throw new IllegalArgumentException("No @Id field found on " +
entity.getClass().getSimpleName());
    }
}
```

4.3 Deadlock-Free Money Transfer Algorithm

To ensure thread safety during transfers, locks must be acquired on both source and destination accounts. Arbitrary lock ordering causes circular wait deadlocks. MiniBank resolves this deterministically.

Listing: Deadlock-Free Transfer via Deterministic Resource Ordering

```
package minibank.service;
import minibank.model.Account;
import minibank.exception.BankException;

public class TransactionProcessor {
    public static void transferDeadlockFree(Account a, Account b, long amount) throws
BankException {
        // Deterministic lock ordering breaks Dijkstra's circular wait condition
        Account first = a.getAccountNumber().compareTo(b.getAccountNumber()) < 0 ? a : b;
        Account second = (first == a) ? b : a;
```

```
        synchronized (first) {
            synchronized (second) {
                a.withdraw(amount);
                b.deposit(amount);
            }
        }
    }
}
```

4.4 Java NIO Audit Log Stream Processor

ReportGenerator uses Java NIO (Files.newDirectoryStream and BufferedReader) to efficiently parse transaction audit logs, calculating aggregate deposits, withdrawals, and net cash flow.

Listing: ReportGenerator: High-Performance Java NIO Log Processor

```
package minibank.service;
import java.io.BufferedReader;
import java.io.IOException;
import java.nio.file.*;
import java.nio.file.attribute.BasicFileAttributes;

public class ReportGenerator {
    public static String generateDailyReport(Path logDir) throws IOException {
        long totalDeposits = 0, totalWithdrawals = 0; int fileCount = 0;
        try (DirectoryStream<Path> stream = Files.newDirectoryStream(logDir, "*.log")) {
            for (Path p : stream) {
                fileCount++;
                try (BufferedReader reader = Files.newBufferedReader(p)) {
                    String line;
                    while ((line = reader.readLine()) != null) {
                        String upper = line.toUpperCase();
                        if (upper.contains("DEPOSIT")) totalDeposits += extractAmount(line);
                        else if (upper.contains("WITHDRAW")) totalWithdrawals +=
extractAmount(line);
                    }
                }
            }
        }
        return String.format("Files: %d | Total Deposits: Rs. %d | Total Withdrawals: Rs. %d |
Net: Rs. %d",
            fileCount, totalDeposits, totalWithdrawals, (totalDeposits - totalWithdrawals));
    }
}
```

Chapter 5: Verification, Testing & Empirical Results

5.1 Automated Unit & Concurrency Test Suite

A rigorous automated test harness, minibank.app.MiniBankTests, was developed to empirically validate system invariants. The test suite covers positive functional paths, exception handling, shortfall computations, and thread concurrency.

Test ID	Test Case Name	Objective & Scenario	Expected Outcome	Status
TC-01	testDepositIncreasesBalance	Verify deposit of Rs. 3000 increments initial balance correctly.	Balance == Initial + 3000	PASSED
TC-02	testWithdrawalOverLimitThrowsException	Attempt withdrawal exceeding balance by Rs. 9,999,999.	Throws InsufficientFundsException with Shortfall	PASSED
TC-03	testNegativeDepositThrowsException	Attempt negative deposit of Rs. -500.	Throws InvalidAmountException	PASSED
TC-04	testConcurrentBatchExecution	Spawn 10 concurrent cross-account transfer worker tasks via ExecutorService.	All 10 tasks execute without deadlock	PASSED

> 📌 **AUTOMATED TEST SUMMARY:** Total Test Cases Executed: 4
> Tests Passed: 4 / 4 (100% Success Rate)
> Execution Time: < 350 ms
> Regression Status: Zero regressions detected across all 14 practical milestones.

5.2 Interactive Console Demonstration & Live Terminal Logs

The interactive CLI provides a clean user interface with 8 options. Below is the verified console transcript captured during execution:

Listing: Actual Console Execution Transcript (Demonstrating All Operations)

```
======================================================
   Welcome to CHARUSAT FTE MiniBank
   Branch: Changa Campus
======================================================
Restored 3 accounts from persistent storage.

------------- MAIN MENU -------------
  1. Open Account
  2. Deposit Funds
  3. Withdraw Funds
```

```
  4. Transfer Funds
  5. View Statement
  6. Generate NIO Audit Report
  7. Run Safe Concurrent Batch
  8. Exit & Persist State

Select an option (1-8): 2
Executing Deposit Rs. 2000 to AC0001...
Deposit successful!

Select an option (1-8): 3
Executing Withdrawal Rs. 1500 from AC0001...
Withdrawal successful!

Select an option (1-8): 5
==================================================
               ACCOUNT STATEMENT
==================================================
Account Number : AC0001
Account Holder : Vaidehi Shah
Account Type   : SavingsAccount
Interest Rate  : 4.0% per annum
Current Balance: Rs. 26000
Status         : ACTIVE
==================================================

Select an option (1-8): 6
==================================================
           MINIBANK END-OF-DAY AUDIT REPORT
==================================================
Log File: transactions.log   | Size: 346 bytes | Modified: 2026-10-28T08:08:51Z
--------------------------------------------------
Total Log Files Scanned: 1
Total Deposited Funds  : Rs. 16000
Total Withdrawn Funds  : Rs. 3000
Net Cash Movement      : Rs. 13000
==================================================

Select an option (1-8): 8
Exiting MiniBank. Persisting state to disk...
Saved 3 accounts to data\accounts.dat
Goodbye!
```

5.3 State Persistence & Data Integrity Verification

To verify that accounts survive system termination, StatePersister was tested across complete JVM shutdown cycles. Before exit, accounts were serialized to data/accounts.dat. Upon subsequent relaunch, the system successfully deserialized all 3 accounts, preserving exact balance states and owner details with 100% data integrity.

Chapter 6: Conclusion, Pedagogical Outcomes & Future Scope

6.1 Project Summary

The MiniBank project successfully implements an end-to-end, multi-tier banking platform while fulfilling all 14 pedagogical milestones prescribed in the CEUE203 course. By combining standalone modular lab stages with a unified enterprise capstone, the repository demonstrates both individual mastery of core concepts and holistic system architecture skills.

6.2 Key Academic & Technical Competencies Acquired

Through this mini-project, the following core engineering skills were developed and reinforced:
• Modern Java Paradigms: Active use of Java 17+ records, pattern matching switch expressions, and lambda expressions.
• Advanced OOP Design: Rigorous application of the SOLID principles—Single Responsibility in services, Open/Closed through account polymorphism, Liskov Substitution across Account hierarchies, and Interface Segregation in transaction rules.
• Metaprogramming with Reflection: Designing custom runtime annotations and building dynamic generic repositories.
• Concurrency Engineering: Managing background thread pools with ExecutorService and applying deterministic lock ordering algorithms to eliminate deadlocks.
• Industrial I/O: Applying binary serialization for state persistence and Java NIO DirectoryStream for high-speed audit log analytics.

6.3 Future Roadmap & Architectural Enhancements

While the current console application is robust and feature-complete, future iterations may incorporate:
1. Relational Persistence: Integrating PostgreSQL with Hibernate/JPA to replace binary serialization with ACID-compliant relational tables.
2. Graphical User Interface (GUI): Developing a modern JavaFX desktop interface or a responsive web dashboard.
3. Microservices & REST API: Wrapping the BankService with Spring Boot to expose secure RESTful banking endpoints.
4. Distributed Transactions: Implementing two-phase commit (2PC) or Saga patterns for distributed cross-bank fund transfers.

References & Submission Metadata

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4. Brian Goetz et al., "Java Concurrency in Practice", Addison-Wesley Professional, 2006.

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Google Form Submission Link: https://forms.gle/qh3yMuouwKyGLdwL6
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